Agentic Travel Planning: Where AI Helped, and Where It Did Not

Stefan Schneider
stefanbschneider@outlook.com

Fabian Missbrenner
Technische Universität Berlin
mail@fabimi.net

Abstract

Many agentic tasks begin with an open-ended goal but are useful only when hard constraints are met. Multi-day travel planning is a clean instance. We describe our entry to the ChinaTravel challenge: a deterministic, environment-grounded planner with an acceptance gate that can only emit itineraries confirmed valid against the environment rules and APIs. AI agents were most valuable in building and improving that system, and selectively useful for language understanding; they did not replace deterministic optimization and verification. The first official held-out submission sharpened this division of labor: grounding generalized, while constraint extraction tuned to one phrasing style did not. Repairing that generalization gap is what the second version contains, and on unfamiliar phrasing it raises the clause satisfaction rate from 75.6 to 89.6. We also report how the system was built: near-continuous, multi-agent "vibe coding" behind a deterministic evaluation harness, which we treat as a result in its own right.

1 Introduction

A growing class of useful tasks asks a system to turn a loosely stated human goal into an output that is not merely plausible but correct. Multi-day travel planning combines fuzzy requirements with exact feasibility. The IJCAI–ECAI 2026 ChinaTravel challenge[1] extends the benchmark of Shao et al. [Shao *et al.*, 2024]: an agent must produce a preference-aligned itinerary over a sandbox of Chinese cities. A request might say:

> "Plan three days in Suzhou, visit the Humble Administrator's Garden between 9 and 11, stay near the old town, avoid taxis, and spend at most CNY 3,000."

The output is structured data, not free text. Hotels, attractions, and restaurants must exist in the sandbox; opening hours, transportation, and costs must be consistent; logical requirements must hold; and commonsense rules such as meal counts must pass. The evaluator measures environmental validity (EPR), logical satisfaction (C-LPR and FPR), and preferences. A single failed hard requirement zeros all preference terms.

The competition has two phases. In the offline first phase, released verifier fields and a generous time budget allowed us to satisfy almost all constraints and then polish preferences. The harder agent phase studied here exposes only query text and environment tools and gives the complete 100-query run one five-hour budget. The agent must infer what to satisfy as well as how to satisfy it.

Our system follows a propose–verify–repair pattern related to LLM-Modulo systems [Kambhampati *et al.*, 2024], but assigns roles asymmetrically: models may interpret language or propose changes; environment APIs establish facts; and deterministic optimization and checks decide what may be emitted.

2 System Design

2.1 Pipeline and trust boundary

Figure 1 follows the example from language to a simplified itinerary. Oracle verifier fields are removed before execution. The only inputs are visible query text and allowed environment tools. Every POI, hotel, restaurant, price, opening hour, and transportation record is read from `WorldEnv`; nothing is generated as a plausible substitute.

2.2 Parsing and normalization

The parser extracts dates, cities, budgets, excluded transportation, named POIs, timed visits, hotel properties, cuisines, and attraction categories. Alternatives remain separate solver variants rather than becoming an impossible conjunction. Normalization resolves names and concepts against each target city's catalog.

Deterministic extraction is fast and testable, but it is not synonymous with understanding. We added phrasing-robust fallbacks for the largest measured recall gaps and an optional Qwen extractor that may fill a missing family but never overwrite a deterministic value. Model-proposed entities and categories are matched against the environment vocabulary before use.

Precision is equally important. Our audit found that in 23–50% of free-prose queries, depending on the set, the parser

[1] https://github.com/LAMDA-NeSy/ChinaTravel

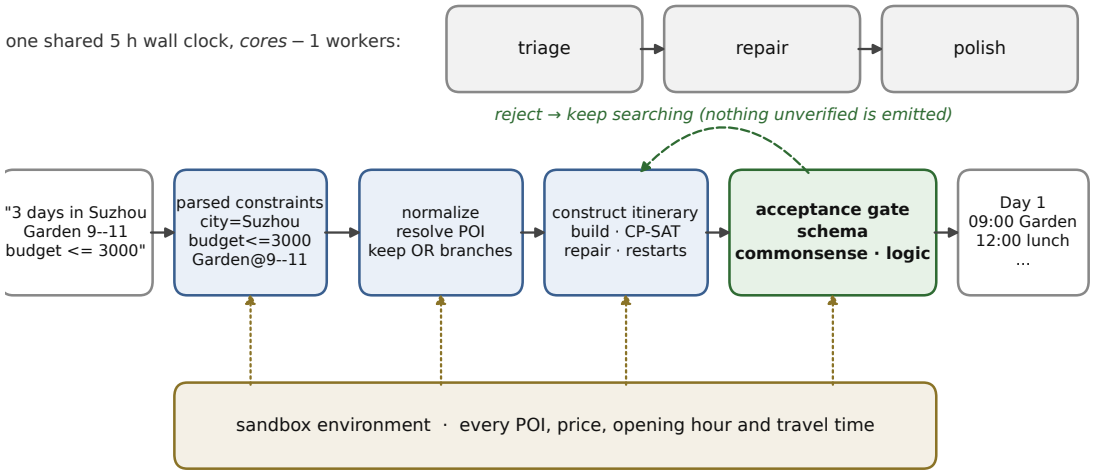


Figure 1: The per-query pipeline and whole-run scheduler, illustrated with the opening example. Optional model-based extraction may fill missing constraints; grounding, construction, repair, and acceptance remain deterministic.

invented a required POI name that did not exist in the target city. Such a query has no feasible solution at all, so it can never earn a full-plan point no matter how long the solver runs. A catalog check now removes only provably unsatisfiable required names and category values, after first trying to salvage a real POI name from the captured text. It reduced this measured failure class to zero on every set we hold.

The most recent recall fix illustrates how narrow these gaps are. Intercity transport was the largest remaining unextracted family on free prose: the extractor recognized "take the train to the destination" but not "we have chosen to fly" or "by high-speed train (G)", missing 33 of 85 such requirements on one released set and 27 of 32 on another. Adding verb-form and qualified-noun patterns, guarded so that a station name or an undecided "fly or take the train" never produces a requirement, removed most of them while leaving precision and the familiar register unchanged.

2.3 Construction, repair, and densification

The deterministic planner constructs a complete base itinerary from grounded candidate pools. It then applies bounded CP-SAT [Perron and Didier, 2024], domain-specific repairs, and seeded restarts. Repairs include placing named activities inside requested windows, inserting absent required POIs, closing transportation gaps, and rebuilding intercity legs. The same acceptance checker evaluates every candidate: query constraints, schema, and commonsense rules must all pass. Accepted candidates are ranked by preference value and cost.

Packing and densification add meals and activities only while the checker continues to pass. A minimal schema-valid itinerary is written before solving so an interrupted run still has an output, but this is availability rather than success: an empty schema-valid itinerary scores zero.

Metric	Familiar 100	Held-out v1	Best held-out
Overall	64.17	33.25	36.41
EPR micro	99.67	99.48	99.64
EPR macro	98.00	95.00	95.00
C-LPR	90.08	55.20	67.77
FPR	44.00	0.00	0.00

Table 1: Local familiar-register baseline, our official v1 result, and the leading team's v1 result on the same held-out set.

2.4 Scheduling the five-hour run

The whole-split scheduler uses *triage*, *repair*, and *polish* passes in parallel across queries. Triage reaches every query cheaply; repair revisits the nearest failures with larger budgets; polish spends remaining time only after hard requirements pass. A keep-best rule makes updates monotonic, and deadline checks prevent starting work that cannot finish within the shared five-hour envelope.

3 Evaluation

3.1 Official feedback

The organizers permit two submitted harness versions and retain the better score. We submitted v1 early to obtain the only available held-out feedback before choosing v2. Table 1 compares that result with the current run on 100 familiar released queries. The columns are diagnostic signals, not identically distributed test sets.

Grounding generalized: held-out micro EPR remained 99.48, within 0.16 of the best entry. Constraint understanding did not: C-LPR fell to 55.20, no query passed every requirement, and all preference terms were therefore zero.

The full field is instructive. Across the five submitted systems, four score a full-plan rate of exactly zero and the fifth scores one point, and every preference term is zero for every team. The held-out queries are evidently dense enough that

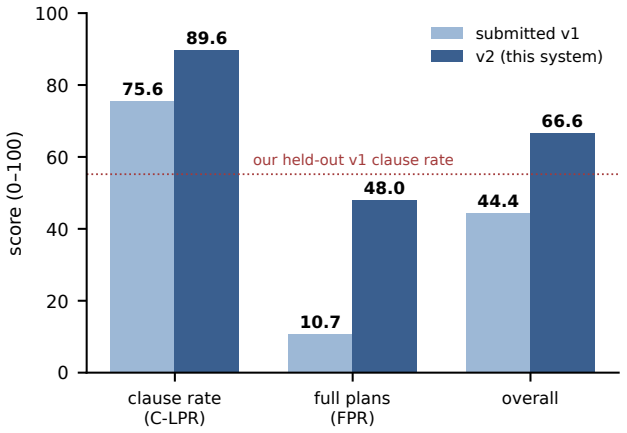


Figure 2: Submitted v1 versus the second version on 150 unfamiliar-phrasing queries, paired at equal budget. The dotted line is our v1 clause rate on the organizers' held-out set, a different query set, shown for scale.

no entrant reliably satisfies all clauses of a single query, so a zero there is a property of the benchmark at this difficulty, not a defect unique to one system. It also means 55% of the metric weight currently pays nobody, and the ranking is decided almost entirely by the per-clause rate at weight 0.25: the entire 3.16-point gap between our v1 and the leading entry is 12.6 clause points. Because the full-plan term is multiplicative in the per-clause rate, it stays near zero until roughly 0.8 and then rises steeply; the first system past that threshold collects the dormant weight in one step.

3.2 What the second version changes

Since the v1 build we repaired the phrasing brittleness described below, removed the unsatisfiable-name failure class, and closed the largest remaining recall gaps. None of that is visible on the familiar register, where the extractor was already complete, so we measured it where it applies: 150 differently phrased released queries, both builds run as a paired experiment at identical budget on one host, scored against an English-literal oracle so the full-plan rate has a real denominator.[2] Figure 2 shows the result: the clause rate rises from 75.6 to 89.6 and the full-plan rate from 10.7 to 48.0. Checked clause by clause, 205 of 1264 clauses move from failing to passing against 27 lost. On the familiar register, where none of this can help, a paired run of the full five-hour pipeline confirms no regression and a small gain: 64.69 to 65.74 overall, with the full-plan rate moving from 45 to 47 of 100 queries.

We deliberately do not convert this into a predicted held-out score. It is one register measured once, and the last prediction we made from a local number was wrong by a factor of two. What it does establish is direction and size in the one metric that currently decides the ranking.

[2]Those released sets state their query text in English but their verifier literals in Chinese, which caps the achievable full-plan rate; we translate the literals with an audited one-to-one map for measurement only, and the agent never reads them.

3.3 Query phrasing

The familiar released set was presented as in distribution, so prioritizing it was a reasonable development choice. Its queries use a regular numbered template, whereas other released sets express similar requirements in varied prose. Offline oracle-backed audits found no missed clauses on the familiar register but 12–29% on the differently phrased sets — we had optimized against the one register we were already perfect on. The precision audit described in Section 2 then revealed the unsatisfiable-name problem, which can pin an otherwise solvable query at a full-plan rate of zero regardless of search time. Both classes of fix are what Figure 2 measures.

3.4 Planning baselines

We ran two small, matched comparisons to identify failure modes. An autonomous tool-using Qwen agent scored 1.88 on eight familiar queries and 2.64 on eight free-prose queries, versus 67.71 and 48.49 for our solver; none of its itineraries passed all commonsense checks. The upstream RuleNeSy baseline [Shao *et al.*, 2024] scored about 9.5 on four queries from each register, grounding most individual facts but still violating at least one commonsense rule in every itinerary. These samples are too small to rank systems, but the failure modes support retaining deterministic acceptance and optimization.

4 Where AI Helped — and Where It Did Not

4.1 The competition as a safe vibe-coding experiment

We used this challenge deliberately as a low-risk environment in which to push agent-driven development much harder than we would on a production system. The task has an objective, automatic scorer; nothing is deployed; and a wrong change is caught by a test or a measurement rather than by an incident. That let us ask a question that is otherwise expensive: how far can coding agents carry a real system if you give them fast, honest feedback and remove yourself from the inner loop?

Very far, as it turned out, given a great deal of scaffolding. We ran multiple paid agent subscriptions across several machines, with one or more agents working almost continuously for eight weeks. The result is 798 commits and 100 merged pull requests over 109 opened ones, a suite of 1339 tests, and 47 committed measurement screens alongside 63 recorded evaluation rows. Humans still made every decision that mattered: what to measure, what a measurement was allowed to prove, and what to submit.

Three practices did the actual work.

Split the roles by cost and by kind of thinking. Expensive reasoning models were used for analysis, diagnosis, and planning: reading measurements, proposing the next experiment, ranking work by expected effect. Faster, cheaper agents then implemented a specified change with tests, and ran evaluations. The planning agent maintained a ranked work list; implementation and evaluation proceeded in parallel branches against it, which also avoided the common failure where an implementing agent silently redefines the task it was given.

Make feedback fast, automatic, and cheap to trust. We invested more in the evaluation harness than in any single planner feature: a paired A/B screen that decides in one or two hours whether a change is worth a long run, a whole-run mode reproducing the submission's exact shape, deterministic seeds, and result ledgers recording every condition needed to decide whether two numbers may be compared. Agents could then evaluate their own work overnight and report a verdict against a criterion written before the run.

Write the invariants down and let them be enforced by tests. Grounding, no oracle leakage, acceptance-before-emission, and flags-off-is-the-floor are recorded as project rules and pinned by tests, so an agent that would violate one fails rather than persuades. Cold-start verification is part of that: it caught a start-up race that would have shipped empty itineraries for an entire split while still exiting successfully, a failure no ordinary local run could expose.

Two limits are worth stating honestly. The volume of agent output is not itself a result: much of it was superseded, and the ratio of committed work to accepted improvement was low. And agents were reliably over-optimistic about their own changes, which is why promotion required an independent measurement rather than an argument.

4.2 Inside the system

Natural-language interpretation is a more nuanced case. LLM-first extraction cut missing free-prose constraint families by 40–65%, but initially converted many misses into values that did not match the environment vocabulary. We therefore connected model output to the same city-specific entity and category catalogs used by the solver. Grounded, fill-only extraction improved one small familiar A/B but was flat on the matched free-prose subset, and still matched fewer clauses than the contemporaneous deterministic parser. The current default consequently makes zero LLM calls, while retaining the grounded extractor as an optional path. This is not a rejection of LLM parsing; it separates recognizing a requirement from resolving its values against the actual database.

The strongest argument for that path is what our deterministic extractor cannot reach. Its largest remaining gap on unfamiliar phrasing is no longer lexical: the queries say "we are more interested in history and culture" or simply "going on a date, please recommend an itinerary". Each implies a database category that no pattern names, and the benchmark's own labels for such phrasings are inconsistent, so hand-written rules would be guessing. Inferring intent is what a language model is for, through the existing tools or an MCP-style interface; its answer only has to be resolved against the closed vocabulary first.

We also tried the obvious hybrid: letting a model repair itineraries the checker rejected, with every proposed edit re-verified before it could be kept. On ten deterministic failures the loop iterated, proposed valid-looking edits, and flipped none of them to passing. It is off by default and reported here as a negative result rather than omitted.

End-to-end itinerary generation was the clearer boundary. Even with environment tools, autonomous Qwen planning accumulated small inconsistencies, and one violation invalidated the whole itinerary. Better prompts and tool interfaces may improve grounding, but dates, opening windows, travel times, budgets, meals, and disjunctions remain a constrained optimization problem. Deterministic solving provides reproducibility, explicit acceptance, and efficient targeted repair.

5 Conclusion

ChinaTravel rewarded a hybrid division of labor. AI agents accelerated the construction and analysis of the system, and model-based extraction recognized language missed by deterministic patterns. Environment tools supplied facts; deterministic search assembled them; and the acceptance gate ensured that only verified itineraries could be emitted. The official v1 result sharpened that boundary: grounding transferred to hidden data, while constraint extraction tuned to one advertised in-distribution phrasing register did not — and repairing that is what the second version measurably contains.

The transferable lesson is methodological. Agentic coding works particularly well in a small, isolated project once objective and fast feedback exists — and this challenge was chosen partly to test that at a scale of continuous multi-agent operation we would not risk on a production system. What made it work was not the agents but the harness around them: written invariants enforced by tests, a measurement contract that says when two numbers may be compared, and a rule that only an independent run may promote a change. Models are valuable at the fuzzy boundary between language and symbols, provided their proposals are grounded. Hard itinerary construction remains optimization.

A production system would replace the sandbox with live map, hotel, rail, flight, and booking APIs or MCP servers. It would also infer user-specific objectives rather than inherit fixed challenge preferences: some travelers want four attractions per day, while others value unplanned time. Live availability and uncertain travel times would require replanning, but the same architecture remains useful. Let AI interpret intent and help build the system; let APIs establish reality; let optimization construct the itinerary; and let deterministic verification decide what is safe to emit.

Acknowledgments

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